Introduction

Prompt engineering involves designing clear instructions that guide Artificial Intelligence systems to generate useful responses. The structure of a prompt can influence the quality, accuracy, and format of the output. By using different prompt styles, users can control how the AI responds.

This section explains four important prompt types: Straightforward Prompts, Tabular Format Prompting, Preceding Question Prompting, and Missing Word Prompting.

Straightforward Prompts

A Straightforward Prompt is the most basic form of prompting. It gives a simple instruction that requires a direct answer. These prompts do not include complex reasoning or multiple tasks. Instead, they request clear and concise information.

For example, the prompt “Define photosynthesis in one sentence.” requires the AI to give a short definition. Photosynthesis is a biological process in which green plants use sunlight, carbon dioxide, and water to produce glucose and release oxygen.

Another prompt example is “List three advantages of electric vehicles.” Electric vehicles offer several advantages including reduced air pollution, lower running costs, and improved energy efficiency. They also help reduce dependence on fossil fuels.

Straightforward prompts are commonly used in quick information retrieval, educational tasks, and simple question-answer systems.

Tabular Format Prompting

Tabular Format Prompting is used when information needs to be presented in an organized structure. Tables allow data to be arranged in rows and columns, making it easier to compare multiple items at once.

For example, the prompt “Compare and contrast AC and DC current in a table.” produces a structured comparison.

Aspect	AC Current	DC Current
Meaning	Alternating Current	Direct Current
Flow of Electricity	Alternates direction	Flows in one direction
Power Source	Power grids	Batteries
Applications	Household appliances	Electronic circuits

Another prompt example is “Provide a table listing five programming languages, their paradigms, and one use case each.”

Language	Paradigm	Use Case
Python	Multi-paradigm	Machine learning
Java	Object-oriented	Enterprise software
C++	Object-oriented	Game development
JavaScript	Event-driven	Web applications
R	Statistical	Data analysis

Tabular prompts improve clarity and help readers quickly identify patterns and comparisons.

Preceding Question Prompting

Preceding Question Prompting begins with a question that introduces the topic. The question encourages the AI to explain the concept in detail.

For instance, the prompt “Why is climate change a global concern? Explain how greenhouse gases contribute to global warming.”

Climate change is a global concern because it affects ecosystems, human health, agriculture, and water resources worldwide. Rising temperatures can cause melting glaciers, rising sea levels, and extreme weather conditions.

Greenhouse gases such as carbon dioxide trap heat in the atmosphere. This trapped heat increases the Earth's average temperature, leading to global warming and long-term climate changes.

Another example prompt is "How do vaccines work? Describe the process of immunization in simple terms."

Vaccines train the immune system to recognize harmful pathogens. They contain weakened or inactive forms of viruses or bacteria. When introduced into the body, the immune system produces antibodies against them. Later, if the real pathogen enters the body, the immune system quickly identifies and destroys it.

This method helps prevent diseases and protects individuals as well as communities.

Missing Word Prompting

Missing Word Prompting involves leaving a blank space in a sentence. The AI must fill in the missing word based on its understanding of the context.

For example, "The capital of France is ____." The correct word that completes the sentence is Paris.

Another example is "In photosynthesis, plants absorb sunlight to produce ____." The correct completion is glucose, which plants use as food.

Missing word prompts are commonly used in educational assessments, quizzes, and language learning systems. They help test knowledge and reinforce understanding of important concepts.

Conclusion

Prompt engineering includes many techniques that help guide AI responses. Straightforward prompts give simple instructions, tabular prompts organize information in structured formats, preceding question prompts encourage detailed explanations, and missing word prompts test knowledge through sentence completion.